

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	
Project Name	HDI Predictor Using Machine Learning
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the project.	Yes	PEP 8 coding style followed.
2	Proper File Structure	Files and folders are logically organized.	Yes	Separate folders for Flask, Model, Dataset, Templates, Static files, Documentation, and Testing.
3	Meaningful Variable Names	Variables clearly describe their purpose.	Yes	Descriptive names improve readability.
4	Function / Method Names	Functions are self-explanatory and meaningful.	Yes	Standard naming conventions followed.
5	Code Comments	Important sections are documented using comments.	Yes	Comments added where necessary for better understanding.
6	Modular Design	Code is divided into reusable modules and functions.	Yes	Flask routes, model loading, and prediction logic are separated.
7	No Redundant Code	Duplicate and unnecessary code removed.	Yes	Optimized for maintainability.
8	Error Handling	Invalid inputs and runtime errors are handled gracefully.	Partial	Basic input validation and exception handling implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	app.py	Python / Flask	Web Application Backend	High
2	HDI.pkl	Scikit-learn / Pickle	Prediction Engine	High
3	index.html	HTML	User Interface	High
4	style.css	CSS	Frontend Styling	High
5	HDI_Prediction.ipynb	Python (Jupyter Notebook)	Model Training & Analysis	Medium

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project structure with clear separation of modules.
Readability	5	Consistent naming conventions and coding standards followed.
Reusability	5	Modular components can be reused for future enhancements.
Documentation / Comments	4	Adequate comments provided; additional documentation can be added if needed.
Overall Score	4.8 / 5	Clean, maintainable, and reusable codebase suitable for deployment and future development.